DUET: Endogenous Failure Detection and Recovery via Dual-Action Manifold Alignment

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***Abstract*— Vision-language-action (VLA) models map vision and language to robot actions. With limited post-training demonstrations, however, they often cannot detect or recover from execution failures that move the robot out of distribution (OOD), such as mis-grasps, dropped objects, unexpected occlusions, and object or robot displacement. Existing solutions rely on external failure detectors and costly intervention or recovery data, leaving detection loosely coupled with control. We introduce DUET (Dual-manifold Unified Execution and Transfer), a dual-action-head module that predicts relative and absolute actions. DUET uses the endogenous disagreement between the two heads as its primary failure signal, complemented by state-distribution distance. The absolute head excludes the current robot state and predicts RVQ-anchored global waypoints from vision and language, providing stable recovery targets when relative control becomes unreliable. We evaluate DUET in LIBERO simulation and real-world manipulation under diverse failure events. DUET preserves nominal success (TODO% vs. TODO% for the relative-only baseline) while achieving TODO% failure recovery success, outperforming the strongest baseline by TODO percentage points.**

I. INTRODUCTION

Vision-language-action (VLA) policies adapt pretrained vision-language models to map visual observations and language instructions directly to robot actions [1], [2], [3]. Despite their semantic generalization, policies post-trained on limited robot demonstrations remain vulnerable to failures during closed-loop execution. We focus on failures that drive the visual observation, robot–object configuration, or induced trajectory out of distribution (OOD) relative to successful demonstrations. They include policy-induced errors, such as an off-center grasp or an accidentally dropped object, as well as external events, such as transient occlusion or unexpected displacement of an object or the robot arm. After such events, a VLA may continue to produce locally plausible actions even though its trajectory is no longer supported by the demonstrations. Detecting these failures and actively recovering from them can therefore substantially improve task success and deployment reliability.

Existing approaches commonly introduce an external monitor, critic, verifier, or predictive model for failure detection [4], and rely on large amounts of human takeover data collected during failed executions to learn recovery [5], [6], [7]. This process is costly because operators must continuously monitor rollouts and provide corrective actions across diverse visual, object, and robot-state failures. Moreover, detection and recovery are learned separately, so a failure alarm is not necessarily aligned with the action needed to recover. We instead seek a single policy that uses endogenous action information to detect and recover from OOD failures without additional failure or intervention data.

DUET takes disagreement between different action spaces as an endogenous signal for failure detection, as illustrated in Fig. 1. In our preliminary experiments, policies trained with relative actions produced smoother and more stable motions during nominal execution, but struggled to recover once a grasp or motion deviated because state-relative commands provide no explicit return target. In contrast, policies trained with absolute actions could redirect the robot toward trajectories supported by the demonstration distribution and retry after a failure. This complementary behavior motivates DUET to learn the same demonstrated behavior with a relative action head and an absolute action head. The relative head serves as the nominal controller, while the absolute head provides global recovery anchors. The two action spaces agree under in-domain execution but necessarily diverge when a failure or OOD event drives the robot away from the demonstrated trajectory, thereby coupling failure detection and recovery through the same signal.

At inference time, DUET converts the predicted absolute waypoint into a correction from the live pose and compares it with the relative action. The two predictions remain consistent along successful trajectories but diverge when a failure drives execution OOD. Their disagreement, together with distance from the training state distribution, determines a soft fusion weight: DUET follows the smooth relative controller during nominal execution, shifts toward the absolute recovery target after failure, and returns to relative control after re-entering the demonstrated support. The same cross-space signal therefore both detects failure and determines the recovery direction.

Our contributions are threefold:

- We propose DUET, a dual-action-head VLA that learns relative and absolute action spaces from successful demonstrations and uses their endogenous disagreement to detect OOD-induced failures without an external detector.
- We design a stateless, RVQ-anchored absolute action head and an adaptive closed-loop controller that converts cross-space disagreement into stable recovery while preserving precise relative control under nominal conditions.
- We validate DUET on diverse simulated and real-world failures, including mis-grasps, object drops, occlusions, and unexpected displacements.

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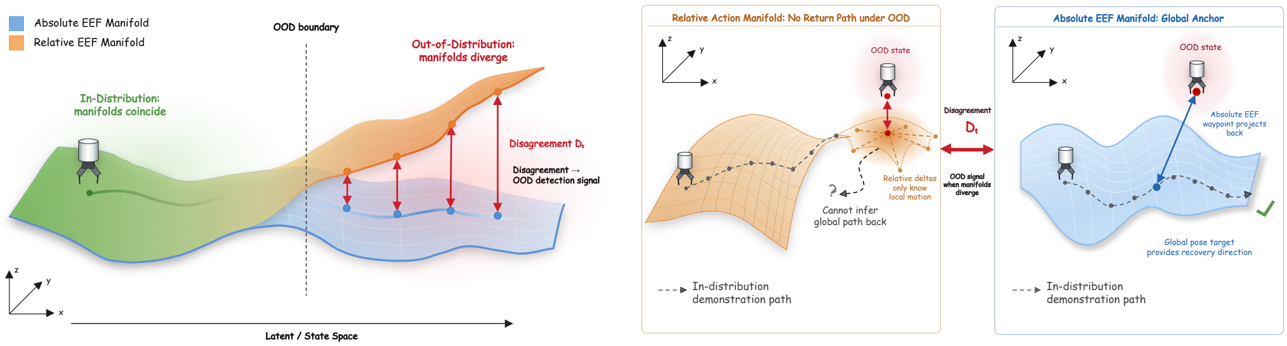

Fig. 1. Dual-manifold failure detection and recovery. The relative and absolute end-effector manifolds agree during nominal execution but diverge when a failure drives the system OOD, exposing an endogenous disagreement score D_t. Relative actions provide precise local motion, whereas absolute waypoints project execution toward a stable demonstrated-manifold anchor.

II. RELATED WORK

A. Failure Detection for Robot Policies

Runtime failure detection estimates whether a policy remains within the support of successful behavior. Diff-DAgger uses diffusion uncertainty to request expert intervention [8], while Sentinel monitors action inconsistency and task progress [9]. FAIL-Detect formulates failure detection as sequential OOD testing using only successful executions [4]; FIPER similarly combines representation novelty with action uncertainty without failure data [10].

VLA-specific methods use internal features or external foundation models. SAFE learns a detector from successful and failed rollouts [11], whereas AHA trains a VLM to detect and diagnose failures [12]. Such methods require additional supervision or a separate detector whose output is not a recovery action. DUET instead detects failures through disagreement between two action spaces learned within the policy, using OOD manifold inconsistency as both an alarm and a recovery signal.

B. Recovery from Execution Failures

Recovery methods commonly expand the training distribution with expert corrections. DAgger queries experts on policy-induced states [5]; intervention-based methods collect human takeovers [13], [6]; and IntervenGen transforms limited interventions into corrective trajectories [7]. These approaches reduce compounding errors but still require additional recovery actions.

Foundation-model systems instead use semantic supervisors. RACER learns from language-annotated recovery trajectories [14], while AHA and VLM controllers diagnose failures and guide downstream correction [12], [15]. Their detection and recovery components remain separately trained or prompted. DUET couples both capabilities without extra recovery data: absolute states from successful demonstrations provide recovery anchors, while cross-space disagreement determines when and how strongly to move toward them.

III. METHOD

A. Problem Formulation

At time t, the robot observes multi-view images o_t, a language instruction l, and its current proprioceptive state s_t. DUET is trained only on successful demonstrations and predicts the demonstrated behavior in two action representations over a horizon K:

$$\hat{\Delta A}_t = f_{\text{rel}}(o_t, l, s_t) \in \mathbb{R}^{K\times 7}, \tag{1}$$

$$\hat{W}_t = f_{\text{abs}}(o_t, l) \in \mathbb{R}^{K\times 7}. \tag{2}$$

Here, ΔA_t contains relative operational-space-control commands and W_t contains future absolute end-effector waypoints $[x, y, z, r_x, r_y, r_z, g]$. We decompose each command as $a_t = [a_{p,t}, a_{g,t}]$, where $a_{p,t} \in \mathbb{R}^6$ controls end-effector pose and $a_{g,t} \in \mathbb{R}$ controls the gripper. A successful trajectory naturally supplies both labels: its recorded delta actions supervise f_{rel}, while its future end-effector states supervise f_{abs}. The geometric analysis and fusion operate on a_p; although the absolute head also predicts gripper state during training, the deployed a_g always comes from the relative head.

B. Dual Action Manifolds

For an instruction l, let $\mathcal{M}_l \subset SE(3)$ denote the end-effector state manifold traced by successful demonstrations, and let $T\mathcal{M}_l$ denote its tangent bundle. Under nominal execution, the relative head predicts a local motion vector

$$\hat{\delta a}_{p,t} = f_{\text{rel}}(o_t, l, s_t)_p \in T_{p_t}\mathcal{M}_l, \tag{3}$$

where p_t is the current end-effector pose. This tangent representation is locally smooth and expressive, but it specifies only how to move from the current chart. Once a failure displaces the robot from the demonstrated support, the extrapolated vector field contains no explicit destination to which the robot should return.

The absolute head instead predicts a future coordinate

$$\hat{w}_{p,t} = f_{\text{abs}}(o_t, l)_p \in \mathcal{M}_l, \tag{4}$$

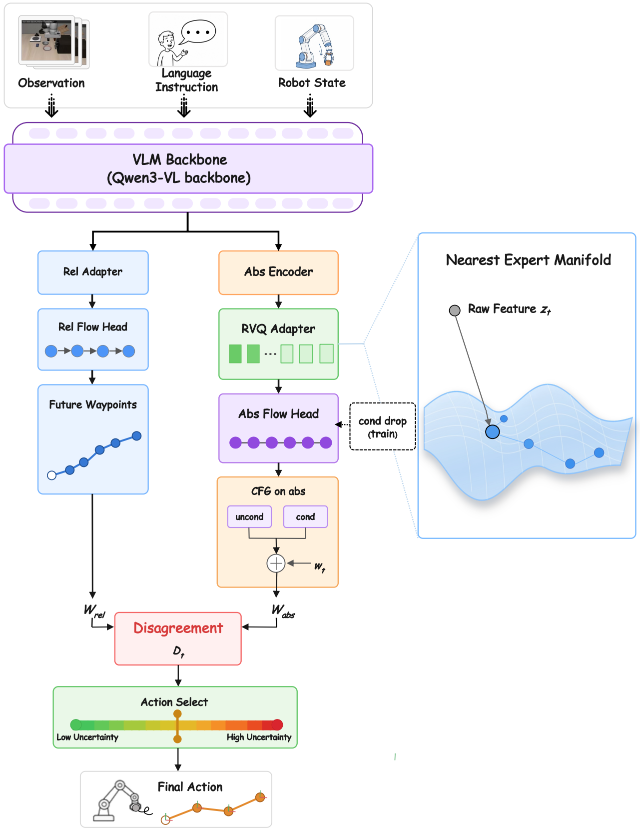

Fig. 2. DUET in four stages: a shared vision-language encoder, relative and absolute action heads, cross-space failure detection, and closed-loop recovery. The relative head provides state-aware local control; the explicitly state-free absolute head uses RVQ-anchored conditions to predict global waypoints. Their disagreement determines the transition between nominal execution and recovery.

which acts as a visual-language-conditioned global anchor. To compare the two representations, we introduce a state-dependent transport map

$$c_t = \Psi_{p_t}(\hat{w}_{p,t}) \in T_{p_t}SE(3), \tag{5}$$

which converts the absolute waypoint into a local correction at the live pose. Along an in-domain trajectory, c_t and $\hat{\delta a}_{p,t}$ describe the same demonstrated motion in two coordinate systems and should therefore align. When a failure drives execution OOD, the relative head extrapolates a local field while the absolute head remains anchored to a demonstrated destination, exposing a cross-space inconsistency.

This geometry couples detection and recovery without asserting a formal OOD guarantee: the norm of the inconsistency provides a scalar failure signal, whereas the transported vector c_t already provides the corresponding recovery direction. DUET can therefore change both when and where the controller moves using the same pair of predictions.

C. Dual-Head Architecture

Figure 2 shows the shared VLM and two flow-matching action heads [16], [17]. The relative head attends to the full hidden sequence and explicitly receives s_t through a numerical state encoder. It is therefore state-aware and serves as the primary controller during nominal execution. The absolute head receives no explicit current proprioception. When state tokens are appended to the causal VLM prompt, they are placed last and excluded from absolute-branch pooling, while the relative head can still attend to them. This separation prevents the absolute predictor from collapsing into an integration of the current state and preserves its role as a global reference.

The absolute branch pools its state-free hidden sequence H^0 and applies residual vector quantization (RVQ) [18], [19]:

$$Z_q = \mathrm{RVQ}\big(\mathrm{Pool}(H^0)\big)\,. \tag{6}$$

RVQ maps continuous conditions to a discrete atlas learned from successful demonstrations, anchoring absolute predictions to recurrent visual-language modes rather than an unconstrained feature vector. Its residual norm is retained only as auxiliary telemetry and is not part of the primary failure score. The absolute flow head additionally uses classifier-free guidance (CFG) [20], enabling its conditional prediction to be softened toward the demonstrated waypoint marginal during recovery.

D. Disagreement-Preserving Learning

The two heads are supervised by the paired labels available in every successful trajectory. Both use flow-matching objectives, while RVQ contributes a commitment loss. The complete default objective is

$$\mathcal{L} = \lambda_{\mathrm{rel}}\mathcal{L}_{\mathrm{rel}} + \lambda_{\mathrm{abs}}\mathcal{L}_{\mathrm{abs}} + \lambda_{\mathrm{vq}}\mathcal{L}_{\mathrm{vq}}, \tag{7}$$

where the absolute labels are normalized future states and the relative labels are the recorded delta commands. We deliberately impose no cross-head consistency loss or distillation: forcing agreement everywhere would suppress the inconsistency needed for failure detection.

DUET instead specializes the branches through asymmetric visual augmentation. The absolute head and RVQ adapter train on both clean and perturbed observations, encouraging the global anchor to remain stable under degraded input. The relative head is optimized only on the clean subset $\mathcal{B}_{\mathrm{clean}}$:

$$\mathcal{L}_{\mathrm{rel}} = \frac{1}{|\mathcal{B}_{\mathrm{clean}}|}\sum_{i\in\mathcal{B}_{\mathrm{clean}}} \ell_{\mathrm{FM}}(\hat{\Delta a}^{(i)}, \Delta a^{(i)}). \tag{8}$$

This gradient truncation preserves relative-head sensitivity while training the absolute branch as the robust reference, making cross-space disagreement more discriminative under visual OOD. For CFG training, the absolute head also replaces its quantized condition with learned null tokens on a subset of samples and learns both conditional and marginal waypoint flows.

E. Joint Failure Detection and Recovery

At deployment, the policy returns both unnormalized action chunks. Let the live pose be $p_t = (x_t, R_t)$ and the pose component of the current absolute waypoint be

$\hat{w}_{p,t} = (\hat{x}_t, \hat{R}_t)$. DUET implements the transport map in Eq. (5) as

$$c_t = \mathrm{clip}\left(\begin{bmatrix}(\hat{x}_t - x_t)/\sigma_p \\ \mathrm{Log}(\hat{R}_t R_t^{-1})/\sigma_R\end{bmatrix}, -1, 1\right), \tag{9}$$

where $\mathrm{Log} : SO(3) \rightarrow \mathfrak{so}(3)$ gives the geodesic rotation correction and σ_p, σ_R convert pose error into action units. The cross-space disagreement is

$$D_t = \frac{\left\| c_t - \mathrm{clip}(\hat{\delta a}_{p,t}, -1, 1) \right\|_2}{\sqrt{6}}. \tag{10}$$

This is the primary failure signal. We supplement it with a coarse state-distance term that measures normalized violation of the training-state bounds:

$$S_t = \mathrm{mean}\left[\frac{\max(s_{\min} - p_t, 0)}{s_{\max} - s_{\min}} + \frac{\max(p_t - s_{\max}, 0)}{s_{\max} - s_{\min}}\right]. \tag{11}$$

The resulting failure score and recovery weight are

$$q_t = D_t + \lambda S_t, \tag{12}$$
$$\alpha_t = \sigma(k(q_t - \tau)). \tag{13}$$

DUET then interpolates between local execution and global recovery while retaining relative gripper control:

$$a_t = [a_{p,t}, a_{g,t}], \tag{14}$$
$$a_{p,t} = (1 - \alpha_t)\hat{\delta a}_{p,t} + \alpha_t c_t, \tag{15}$$
$$a_{g,t} = \hat{\delta a}_{g,t}, \tag{16}$$

where $[\hat{\delta a}_{p,t}, \hat{\delta a}_{g,t}]$ is the relative-head command. Thus relative control dominates along the demonstrated support, while a rising failure score shifts a_p toward the absolute recovery direction; control returns smoothly after the two representations realign. When requesting a new chunk, the previous fusion weight also softens the absolute-head CFG scale,

$$w_t = w_{\max} - (w_{\max} - w_{\min})\alpha_{t-1}. \tag{17}$$

The disagreement D_t therefore determines *when* recovery is needed, while the vector c_t determines *how* to recover.

IV. EXPERIMENTS

A. Benchmark and Data

We evaluate on LIBERO [21], using the StarVLA LeRobot data pipeline. The dual-label data configuration uses current plus future state indices $0, \ldots, 8$, while action labels use the native delta OSC actions. The default draft configuration trains on the `libero_10_hybrid` mixture, with planned extensions to `libero_all_hybrid`. For few-shot experiments, the data loader limits demonstrations per task through `max_demos_per_task`.

TODO: Insert gate response plot.
Expected curves: injected perturbation interval, disagreement D_t, state distance S_t, fusion weight α_t, and CFG scale w_t over time.

Fig. 3. Planned closed-loop telemetry visualization for perturbation recovery experiments.

TABLE I
LIBERO NOMINAL AND FAILURE RECOVERY SUCCESS RATES. ALL QUANTITATIVE ENTRIES ARE TODO PLACEHOLDERS AND MUST BE FILLED AFTER EVALUATION.

Method	Demos/task	Clean SR	Failure Recovery SR
Relative-only VLA	TODO	TODO	TODO
Absolute-only waypoint	TODO	TODO	TODO
Fixed fusion, $\alpha = 0.5$	TODO	TODO	TODO
DUET adaptive	TODO	TODO	TODO

B. Implementation Details

DUET is implemented as `QwenGR00THybrid` in StarVLA. The backbone is Qwen-VL with hidden dimension 2048. The relative branch uses a 16-layer DiT-B flow-matching head. The absolute branch uses an 8-layer DiT-B flow-matching head with 16 RVQ condition tokens of output dimension 1024. RVQ uses three residual quantizers with codebook size 512 and code dimension 256. We use visual augmentation probability $p_{\mathrm{aug}} = 0.3$, CFG condition-dropout probability $p_{\mathrm{drop}} = 0.15$, and disable the optional learned gate by default. Training uses AdamW and the schedule specified in the repository configuration.

C. Failure Detection and Recovery Protocol

We evaluate both policy-induced and externally induced failures. The former include off-center grasps and accidental object drops; the latter include camera occlusion and unexpected displacement of objects or the robot arm. For controlled simulation, we additionally inject action-space perturbations with fixed start time, duration, and magnitude. We record detection accuracy and latency, recovery success and time, action traces, dual-branch disagreement, state-distance score, fusion weight, and CFG scale.

D. Main Results

Table I will report nominal and failure recovery success rates. Relative-only control should remain competitive during nominal execution but be less robust after failures, while DUET should improve recovery by increasing the manifold-projection weight only when needed.

E. Ablations

The ablations in Table II isolate condition anchoring, guidance adaptation, branch specialization, state prompt injection, adaptive fusion, and an optional learned gate trained with synthetic state perturbations. Additional metrics include failure-detection accuracy and latency, mean recovery time, maximum displacement after failure, and area under the α_t curve.

TABLE II
PLANNED ABLATIONS. TODO ENTRIES SHOULD BE FILLED WITH SUCCESS RATE, RECOVERY RATE, AND TELEMETRY STATISTICS.

Ablation	Clean SR	Failure Recovery SR
DUET full	TODO	TODO
No RVQ anchoring	TODO	TODO
No CFG adaptation	TODO	TODO
No relative gradient truncation	TODO	TODO
State-to-VLM disabled	TODO	TODO
DUET + learned gate	TODO	TODO
Fixed $\alpha = 0$	TODO	TODO
Fixed $\alpha = 1$	TODO	TODO

V. LIMITATIONS

DUET assumes access to absolute state labels aligned with the relative action chunks. This is natural in LIBERO and many robot logs, but may require additional calibration or state estimation on real hardware. The closed-loop gate currently uses simple min-max state statistics as a coarse proxy for distance to the demonstrated state manifold, which may be insufficient for highly multi-modal workspaces. The absolute branch is deliberately stateless; while this supports recovery, it may reduce waypoint accuracy in tasks where the current robot state contains task-relevant information not recoverable from images. Finally, all quantitative claims remain pending until the planned simulation and real-world failure experiments are completed.

VI. CONCLUSION

We presented DUET, a dual-branch VLA policy for endogenous failure detection and recovery. By aligning a local relative-action tangent field with a stateless, RVQ-anchored absolute-waypoint manifold, DUET uses cross-space disagreement to detect when execution becomes OOD and to activate a closed-loop recovery direction. This unifies failure detection and recovery without an external detector or additional recovery demonstrations.

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REFERENCES

[1] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski, T. Ding, D. Driess, A. Dubey, C. Finn, P. Florence, C. Fu, M. G. Arenas, K. Gopalakrishnan, K. Han, K. Hausman, A. Herzog, J. Hsu, B. Ichter, A. Irpan, N. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, I. Leal, K.-H. Lee, S. Levine, Y. Lu, H. Michalewski, I. Mordatch, K. Pertsch, K. Rao, K. Reymann, M. Ryoo, G. Salazar, P. Sanketi, P. Sermanet, J. Singh, A. Singh, R. Soricut, H. Tran, V. Vanhoucke, Q. Vuong, A. Wahid, S. Welker, P. Wohlhart, J. Wu, F. Xia, T. Xiao, P. Xu, S. Xu, T. Yu, and B. Zitkovich, "RT-2: Vision-language-action models transfer web knowledge to robotic control," *arXiv preprint arXiv:2307.15818*, 2023.

[2] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, T. Lampe, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, C. Finn, and P. Liang, "OpenVLA: An open-source vision-language-action model," *arXiv preprint arXiv:2406.09246*, 2024.

[3] StarVLA Community, "StarVLA: A lego-like codebase for vision-language-action model developing," https://github.com/starVLA/starVLA, 2026.

[4] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez, P. Miller, R. Lee, P. Shah, R. Ambrus, H. Nishimura, and M. Itkina, "Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies," in *Proceedings of Robotics: Science and Systems*, 2025.

[5] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, 2011.

[6] H. Liu, S. Nasiriany, L. Zhang, Z. Bao, and Y. Zhu, "Robot learning on the job: Human-in-the-loop autonomy and learning during deployment," in *Proceedings of Robotics: Science and Systems*, 2023.

[7] R. Hoque, A. Mandlekar, C. R. Garrett, K. Goldberg, and D. Fox, "IntervenGen: Interventional data generation for robust and data-efficient robot imitation learning," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2024, pp. 2840–2846.

[8] S.-W. Lee, X. Kang, and Y.-L. Kuo, "Diff-DAgger: Uncertainty estimation with diffusion policy for robotic manipulation," in *IEEE International Conference on Robotics and Automation*, 2025.

[9] C. Agia, R. Sinha, J. Yang, Z. Cao, R. Antonova, M. Pavone, and J. Bohg, "Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress," in *Proceedings of the Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 689–723.

[10] R. Römer, A. Kobras, L. Worbis, and A. P. Schoellig, "Failure prediction at runtime for generative robot policies," in *Advances in Neural Information Processing Systems*, 2025.

[11] Q. Gu, Y. Ju, S. Sun, I. Gilitschenski, H. Nishimura, M. Itkina, and F. Shkurti, "SAFE: Multitask failure detection for vision-language-action models," in *Advances in Neural Information Processing Systems*, vol. 38, 2025.

[12] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang, S. Tian, W. Yuan, R. Krishna, D. Fox, A. Mandlekar, and Y. Guo, "AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation," in *International Conference on Learning Representations*, 2025.

[13] J. Spencer, S. Choudhury, M. Barnes, M. Schmittle, M. Chiang, P. Ramadge, and S. Srinivasa, "Learning from interventions: Human-robot interaction as both explicit and implicit feedback," in *Proceedings of Robotics: Science and Systems*, 2020.

[14] Y. Dai, J. Lee, N. Fazeli, and J. Chai, "RACER: Rich language-guided failure recovery policies for imitation learning," in *IEEE International Conference on Robotics and Automation*, 2025, pp. 15 657–15 664.

[15] H. Chen, Y. Yao, R. Liu, C. Liu, and J. Ichnowski, "Automating robot failure recovery using vision-language models with optimized prompts," *arXiv preprint arXiv:2409.03966*, 2024.

[16] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *The International Journal of Robotics Research*, 2024.

[17] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, J. Jakubczak, T. Jones, A. Khazatsky, S. Levine, A. Li-Bell, Y. Lu, S. Nair, K. Pertsch, Q. V. Xu, F. Xia, and T. Xiao, "π_0: A vision-language-action flow model for general robot control," *arXiv preprint arXiv:2410.24164*, 2024.

[18] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, "Neural discrete representation learning," in *Advances in Neural Information Processing Systems*, 2017.

[19] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, "Soundstream: An end-to-end neural audio codec," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2022.

[20] J. Ho and T. Salimans, "Classifier-free diffusion guidance," *arXiv preprint arXiv:2207.12598*, 2022.

[21] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, "LIBERO: Benchmarking knowledge transfer for lifelong robot learning," in *Advances in Neural Information Processing Systems*, 2023.